

backend/core/distillation.py Documentation (Roman Urdu)

Yeh file application ke **Phase 3 (Context Distillation Engine)** ka heavy-duty brain hai. Jab user RAG (Retrieval-Augmented Generation) use kar ke PDF ya website par sawal poochta hai, toh system document se chunk (hissay) uthata hai. Mgar yeh chunks noisy aur faltu alfaaz se bharay ho skty hain jo model ko confuse kreingy aur limit ziada kr dein gy. Yeh engine "Distillation" (khalis panna) lata ha.

File Ka Maqsad (Purpose)

Isme AI aur Data Structure k asool use kiye gae hain taake bade se bada text compress ho, naya refined response template banay, aur multi-step retrieval (khoj) mumkin banaye.

Code Blocks aur Classes ki Tafseel

1. class ContextCompressor

- **Kya karta hai?:** Extra formatting (lines, spacing), legal boilerplates ("all rights reserved") remove karta hai. Aur sentence level par `_query_filter` ke zariye sirf woi lines rakhta h jo direct query mein puche gae lafzo (terms) se match khayein.

2. class DeduplicationEngine

- **Kya karta hai?:** Similar (Duplicate) data blocks ko identify karna.
- **Logic:** Text string ki `_fingerprint()` (hashes) banata aur math k `_jaccard` logic se similiarity (misal 85%) check karta hai. Agar ek jaisa paragraph dubara mil raha ho vector se, usko delete kardeta ha.

3. class PromptTemplateEngine

- **Kya karta hai?:** User ka sawal decide kar k Prompt adjust karna.
- **Logic:** Agar user ne "code", "bug" pucha hai toh yeh automatically `TEMPLATES["code"]` (code analysis format) lagy gha, agar history sa munasibat jani hai toh `analytical` wala format prompt system generate kary ga (ta k output structured aiw).

4. `class QueryComposer` aur `MultiHopRetriever`

- **Composer:** Bade (complex) sawalo me jahan user "and" ya "also" laga kar 2 baaten puche, yeh split marke 2 sub-queries process karta ha baghir LLM involve kiye.
- **MultiHop:** Ek results layer aanay k baad us text mein se Noun aur Names (Entities) nikal kar ek doosra hop (search) engine pr dal deta ha.

5. `class DistillationPipeline` (Orchestrator)

- **Kya karta hai?:** In sb bikhray hue tools ka aik controller orchestrator (conductor).
- **Kaam ki tarhterb (Flow):** Decompose -> Retrieve chunks -> Remove logic -> Deduplicate -> Score Confidences -> Prompt Assemble -> Send!